

# Qatar-1b

R-2761

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_250628 · created 2026-09-03T00:24:32+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460854.8635 +/- 0.0025 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1909 +/- 0.0097               |
| Transit depth (Rp/Rs)^2                           | 3.65 +/- 0.37 [%]               |
| Orbital Inclination (inc)                         | 89.05 +/- 1.11                  |
| Airmass coefficient 1 (a1)                        | 1.753 +/- 0.036                 |
| Airmass coefficient 2 (a2)                        | -0.45 +/- 0.14                  |
| Scatter in the residuals of the lightcurve fit is | 2.09 %                          |
| Best Comparison Star                              | #1 - [497, 317]                 |
| Optimal Aperture                                  | 3.98                            |
| Optimal Annulus                                   | 13.29                           |
| Transit Duration (day)                            | 0.0859 +/- 0.0012               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                         |
|--------------------------------------|---------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1909 $\pm$ 0.0097 vs 0.1463 (deviation 4.6 $\sigma$ ) |
| Duration fit vs published            | 0.0859 d vs 0.0692 d (deviation 13.94 $\sigma$ )        |
| Mid-time O-C                         | 1.2 min                                                 |
| Depth SNR                            | 19.7                                                    |
| Residual scatter                     | 2.09 %                                                  |

### Light curve

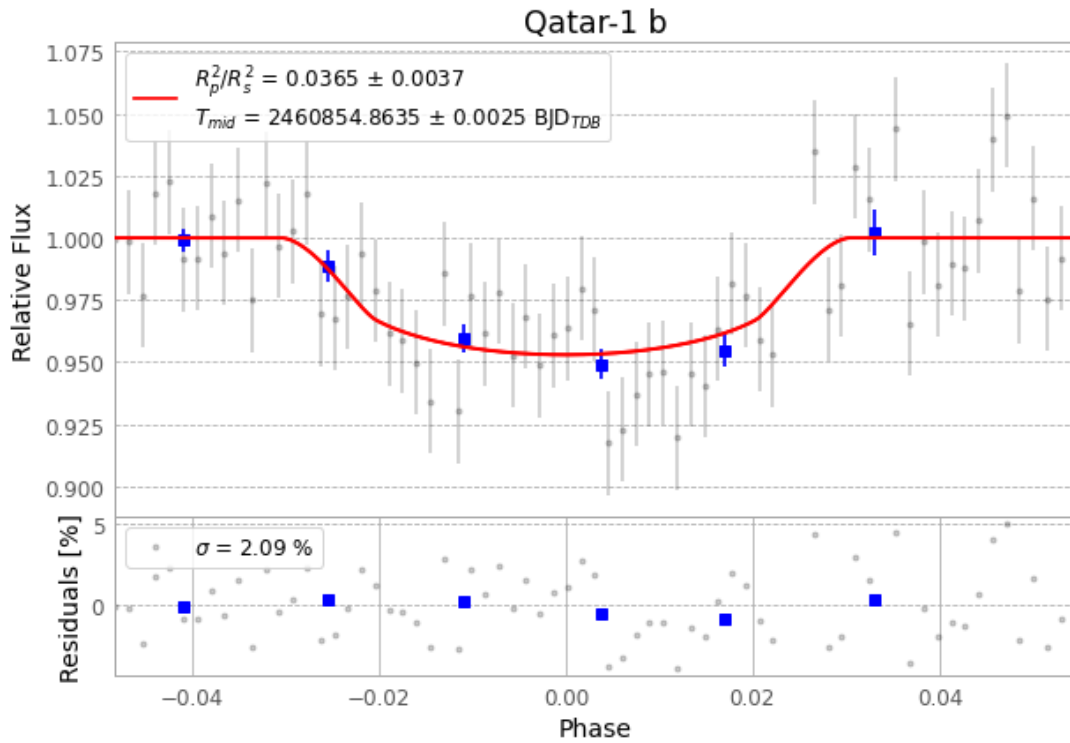


Figure 1 — FinalLightCurve\_Qatar-1 b\_2025-06-28.png (SHA-256 cf9feace3088f6c2...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 236], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 56a112facff5812f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

71 FITS frames (47 MB), Qatar-1250628070317.FITS → Qatar-1250628103316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250628.log (2fd8d87b9da5...), FinalLightCurve\_Qatar-1 b\_2025-06-28.png (cf9feace3088...), FinalLightCurve\_Qatar-1 b\_2025-06-28.csv (8ef6593cedf1...)

## Compute resources

Wall time 161.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:24Z.